

```

10REM > $.!Survey.!RunImage
20
30REM Bart, November 1989 v1.30
40
50REM This Cave Surveying program is based on Waddingtons SU_BBC / SU_1/65
60
70ON ERROR MODE 12 : REPORT : PRINT" at line ";ERL : END
80
90PROCinitmain
100PROCinitsurvey
110PROCinitprint
120PROCdisplay
130CLS
140END
150
160DEF PROCinitmain
170REM Station data structure
180maxstations% = 1500
190stationsize% = 40
200s_name%      = 0
210s_flags%     = 8
220s_tocount%   = 12
230s_fromcount% = 16
240s_x%         = 20
250s_y%         = 24
260s_z%         = 28
270s_w%         = 32
280s_h%         = 36
290
300mo_bit%      = %10000000 :REM Mole Phone
310vz_bit%      = %01000000 :REM Visited
320or_bit%      = %00100000 :REM Origin
330cl_bit%      = %00010000 :REM Part of a closure loop
340
350REM Vector data structure
360maxvectors%  = 1500
370vectorsize%  = 32
380v_from%      = 0
390v_to%        = 4
400v_length%    = 8
410v_bearing%   = 12
420v_incline%   = 16
430v_dx%        = 20
440v_dy%        = 24
450v_dz%        = 28
460
470REM Information on survey
480maxsurveys%  = 50
490surveysize%  = 170
500survey_no%   = 0           :REM The number of surveys, the north pointer
510x_north%     = 4           :REM position and the survey rotation just
520y_north%     = 8           :REM appear once in the information
530su_rot%      = 12          :REM Along with the extent of the whole system
540su_mag%      = 16          :REM magnification
550su_sta%      = 20          :REM Total number of stations
560su_vec%      = 24          :REM and vectors
570x_min%       = 28          :REM Extent of whole system
580y_min%       = 32
590z_min%       = 36
600x_max%       = 40
610y_max%       = 44
620z_max%       = 48
630total_len%   = 52          :REM total length of all merged surveys
640
650station_no%  = 0           :REM Each survey has number of stations
660vector_no%   = 4           :REM number of vectors
670start_sta%   = 8           :REM Start position in main station list
680start_vec%   = 12
690survey_len%  = 16          :REM Along with its length
700title%       = 20          :REM title length currently 50 bytes
710people%      = 70          :REM list of surveyors

```

```

720date%      = 120      :REM date of survey
730
740DIM stations% maxstations%*stationsize%, slist%(maxstations%, 3)
750DIM vectors% maxvectors%*vectorsize%, pt%(maxvectors%, 3)
760DIM info% maxsurveys%*surveysize%
770
780Diving      = 0      :REM Length, bearing, absolute depth
790Normal      = 1      :REM Length, bearing, inclination
800UpDown      = 2      :REM Length, bearing, relative depth
810
820Free        = 0      :REM Formats
830Fixed       = 1
840
850Raw         = 0      :REM Input conversion flags
860Units       = 1
870Internal    = 2
880
890filepath$ = "<Survey$Dir>.SurveyData."
900filename$ = ""
910
920maxfiles% = 4
930DIM handle%(maxfiles%), file$(maxfiles%)
940eptr% = 0
950eflgs% = 0
960
970DIM buf% 255, tmp%(50) :REM Some storage
980CR = 13
990t1% = 0      :REM Some global temporary vars
1000t2% = 0
1010ENDPROC
1020
1030DEF PROCinitsurvey
1040info%!survey_no% = 0
1050info%!x_north% = 0
1060info%!y_north% = 0
1070info%!su_rot% = 0
1080info%!su_sta% = 0
1090info%!su_vec% = 0
1100info%!su_mag% = 100 :REM Full size
1110info%!x_min% = 1E6 :REM extent of cave system
1120info%!y_min% = 1E6
1130info%!z_min% = 1E6
1140info%!x_max% = -1E6
1150info%!y_max% = -1E6
1160info%!z_max% = -1E6
1170info%!total_len% = 0
1180info%!(surveysize%+station_no%) = 0
1190info%!(surveysize%+vector_no%) = 0
1200info%!(surveysize%+start_sta%) = 0
1210info%!(surveysize%+start_vec%) = 0
1220info%!(surveysize%+survey_len%) = 0
1230$(info%+surveysize%+title%) = ""
1240$(info%+surveysize%+people%) = ""
1250$(info%+surveysize%+date%) = ""
1260style% = Normal
1270format% = Free
1280depth = 0.0 :REM current depth
1290unit_factor = 1.0 :REM default to metres
1300factor% = 10 :REM to convert to internal format
1310compass_adj = 0.0 :REM correction for compass bearings
1320clino_adj = 0.0 :REM and clinometer readings
1330surface_depth = 0.0 :REM initial error on depth gauge
1340gauge_factor = 1.0 :REM and correction factor
1350old_to$ = "" :REM Previous survey station
1360leg$ = "" :REM station at start of leg
1370leg% = 0 :REM number stations in this leg
1380handle%() = 0
1390files% = 0
1400survey% = FALSE
1410st_no% = 0 :REM station number
1420st_name$ = "" :REM station name

```

```

1430st_base% = 0
1440su_no% = 0          :REM survey number
1450ENDPROC
1460
1470DEF PROCinitdisplay
1480OFF
1490PROCtrig
1500*WimpPalette <Survey$Dir>.!Palette
1510*Pointer 1
1520help$ = ""
1530oldhelp$ = ""
1540help_on% = TRUE
1550mx% = 0 : my% = 0 : mb% = 0
1560wxi%=0      : wyi%=0
1570vllx%=0      : vby%=0      : vrax%=959 : vty%=959
1580wlx%=1E6     : why%=1E6     : wrax%=-1E6 : wty%=-1E6
1590mlx%=vrax%   : mby%=0       : mrx%=1279 : mty%=vty%
1600orgx% = 0 : orgy% = 0
1610VDU 23, 255, 255,255,255,255,255,255,255,255
1620bank% = 1
1630Plan = 0
1640Side = 1
1650Elev = 2
1660view% = Plan
1670grid% = FALSE
1680quit% = FALSE
1690e_bit% = 1
1700s_bit% = 2
1710i_bit% = 4
1720p_bit% = 8
1730*Fx 4 1
1740*Fx 112, 2
1750PROCinitnorth
1760PROCiniticons
1770PROCshow(st_no%,wlx%,wby%,wrx%,wty%, e_bit% OR s_bit% OR i_bit% OR p_bit%)
1780*Fx 112, 1
1790PROCshow(st_no%,wlx%,wby%,wrx%,wty%, e_bit% OR s_bit% OR i_bit% OR p_bit%)
1800ENDPROC
1810
1820DEF PROCinitnorth
1830RESTORE 1910
1840READ Npts%
1850DIM north%(Npts%, 3)
1860FOR i% = 1 TO Npts%
1870   FOR j% = 0 TO 3
1880     READ north%(i%, j%)
1890   NEXT
1900NEXT
1910DATA 7
1920DATA 32,0, 32,96, 0,64, 32,96, 64,64, 32,96, 16,32, 48,32 :REM arrow
1930DATA 16,112, 16,144, 16,144, 48,112, 48,112, 48,144 :REM N
1940ENDPROC
1950
1960DEF PROCiniticons
1970LOCAL icon$
1980maxicons% = 35
1990DIM icon$(maxicons%, 1)
2000RESTORE 2150
2010iw% = 80
2020ih% = 80
2030icons% = -1
2040READ icon$, help$
2050WHILE icon$ <> "!"
2060   icons% += 1
2070   icon$(icons%, 0) = icon$
2080   IF help$ = "" THEN help$ = icon$(icons%-1, 1)
2090   icon$(icons%, 1) = help$
2100   READ icon$, help$
2110ENDWHILE
2120ix% = mlx%
2130iy% = mty%+4

```

```

2140
2150DATA new,      " Read a new survey "
2160DATA load,     " Load existing  "
2170DATA save,     "Save current survey"
2180DATA merge,    " Merge survey"
2190
2200DATA grid,     " Toggle OS grid"
2210DATA north,    "Move North Pointer"
2220DATA swap,     " Swap survey"
2230DATA command,  " Use command line"
2240
2250DATA data1,    " Print Data"
2260DATA paper,    " Change paper size"
2270DATA print,    " Print survey"
2280DATA quit,     " Exit program"
2290
2300DATA next,     "Next survey station"
2310DATA slef,     " Station name"
2320DATA sright,   ""
2330DATA prev,     " Previous station"
2340
2350DATA view,     " Change view"
2360DATA zoomin,   " Reduce Survey"
2370DATA magnify,  " Set Magnification"
2380DATA zoomout,  " Magnify Survey"
2390
2400DATA upleft,   " Position controls"
2410DATA up,       ""
2420DATA upright,  ""
2430DATA rotright, "Change orientation"
2440DATA left,     " Position controls"
2450DATA help,     "Toggle help on/off"
2460DATA right,    " Position controls"
2470DATA degree,   "Bearing from North"
2480DATA downleft, " Position controls"
2490DATA down,     ""
2500DATA downright,""
2510DATA rotleft,  "Change Orientation"
2520DATA !,        ""
2530ENDPROC
2540
2550DEF PROCinitprint
2560DIM wnd% 16, mat% 16, org% 16, blk% 16
2570printfile$ = "printer:"
2580papersize% = 4
2590OS_Scale% = 50000
2600SYS "OS_SWINumberFromString",, "PDriver_SelectJob" TO selectjob%
2610SYS "OS_SWINumberFromString",, "PDriver_AbortJob" TO abortjob%
2620SYS "OS_SWINumberFromString",, "PDriver_GiveRectangle" TO giverect%
2630SYS "OS_SWINumberFromString",, "PDriver_DrawPage" TO drawpage%
2640SYS "OS_SWINumberFromString",, "PDriver_GetRectangle" TO getrect%
2650SYS "OS_SWINumberFromString",, "PDriver_EndJob" TO endjob%
2660SYS "PDriver_PageSize" TO , p_x%, p_y%, p_l%, p_b%, p_r%, p_t%
2670PROCsetblock(wnd%, 0, -50, 1279, 1799)
2680PROCsetblock(mat%, 1<<16, 0, 0, 1<<16)
2690PROCsetblock(org%, p_l%, p_b%+72000, 0, 0)
2700ENDPROC
2710
2720DEF PROCdisplay
2730PROCinitdisplay
2740REPEAT
2750 PROCmouse(0)
2760 IF FNonicons(mx%, my%) THEN
2770 icon% = FNiconnumber(mx%, my%)
2780 PROChelp(icon$(icon%, 1), 12, 13)
2790 IF mb% > 0 THEN
2800 REM Peter
2810 REM flags% = EVAL("FN"+icon$(icon%, 0))
2820 REM GOTO 3090 ? just after ENDCASE !
2830 CASE icon$(icon%, 0) OF
2840 WHEN "new" : flags% = FNnew

```

```

2850      WHEN "load"      : flags% = FNload
2860      WHEN "save"      : flags% = FNsave
2870      WHEN "merge"     : flags% = FNmerge
2880      WHEN "grid"      : flags% = FNgrid
2890      WHEN "north"     : flags% = FNnorth
2900      WHEN "swap"      : flags% = FNswap
2910      WHEN "command"   : flags% = FNcommand
2920      WHEN "data1"     : flags% = FNdata1
2930      WHEN "paper"     : flags% = FNpaper
2940      WHEN "print"     : flags% = FNprint
2950      WHEN "quit"      : flags% = FNquit
2960      WHEN "upleft"    : flags% = FNupleft
2970      WHEN "up"        : flags% = FNup
2980      WHEN "upright"   : flags% = FNupright
2990      WHEN "rotright"  : flags% = FNrotright
3000      WHEN "left"     : flags% = FNleft
3010      WHEN "help"     : flags% = FNhelp
3020      WHEN "right"    : flags% = FNright
3030      WHEN "degree"   : flags% = FNdegree
3040      WHEN "downleft" : flags% = FNdownleft
3050      WHEN "down"     : flags% = FNdown
3060      WHEN "downright" : flags% = FNdownright
3070      WHEN "rotleft"   : flags% = FNrotleft
3080      WHEN "view"     : flags% = FNview
3090      WHEN "zoomin"    : flags% = FNzoomin
3100      WHEN "magnify"   : flags% = FNmagnify
3110      WHEN "zoomout"   : flags% = FNzoomout
3120      WHEN "next"     : flags% = FNnext
3130      WHEN "sleft"    : flags% = FNsleft
3140      WHEN "sright"   : flags% = FNsright
3150      WHEN "prev"     : flags% = FNprev
3160      ENDCASE
3170      IF flags% > 0 THEN PROCshow(st_no%, wlx%, wby%, wrx%, wty%, flags%)
3180      ENDIF
3190  ELSE
3200      IF FNinwindow(mx%, my%) THEN
3210          PROChelp(" Display window", 12, 13)
3220          PROCmouse(4)
3230          PROCshow(st_no%, wlx%, wby%, wrx%, wty%, p_bit%)
3240      ELSE
3250          IF FNonscale(mx%, my%) AND mb% > 0 THEN
3260              PROCscale
3270          ELSE
3280              PROChelp(" Information area", 12, 13)
3290          ENDIF
3300      ENDIF
3310  ENDIF
3320 UNTIL quit%
3330 ENDPROC
3340
3350 DEF FNinwindow(x%, y%)
3360 = x% > vlx% AND x% < vrx% AND y% > vby% AND y% < vty%
3370
3380 DEF FNonicons(x%, y%)
3390 = x% > ix% AND x% < (ix% + 4*iw%) AND y% > (iy% - ((icons% + 1) DIV 4)*ih%) AND y% < iy%
3400
3410 DEF FNonscale(x%, y%)
3420 = x% > ix% AND x% < (ix% + 4*iw%) AND y% > ih% AND y% < ih%*2
3430
3440 DEF FNiconnumber(x%, y%)
3450 = ((iy% - y%) DIV ih%)*4 + (x% - ix%) DIV iw%
3460
3470 DEF PROChelp(help$, fc%, bc%)
3480 IF help_on% THEN
3490     IF help$ <> oldhelp$ THEN
3500         oldhelp$ = help$
3510         PROCiconpos(45, 13, t1%, t2%)
3520         PROCupdate(t1% - 8, t2%, help$, 19, fc%, bc%)
3530     ENDIF
3540 ENDIF
3550 ENDPROC

```

```

3560
3570DEF PROCscale
3580PROCiconpos(43, 9, t1%, t2%)
3590OS_Scale% = VAL(FNginput(t1%, t2%, STR$(OS_Scale%), 6, 7, 0))
3600PROCiconpos(41, 13, t1%, t2%)
3610PROCupdate(t1%, t2%, "OS Scale 1:" + STR$(OS_Scale%), 18, 12, 13)
3620ENDPROC
3630
3640DEF FNnew
3650filename$ = FNginput(mlx%+96, 1023-16, filename$, 10, 7, 0)
3660PROCcleardisplay
3670PROCreadsurvey(filepath$, filename$)
3680VDU 26
3690= s_bit% OR i_bit% OR p_bit%
3700
3710DEF FNload
3720PROCinitsurvey
3730filename$ = FNginput(mlx%+96, 1023-16, filename$, 10, 7, 0)
3740survey% = FNloadsurvey(filepath$, filename$)
3750IF survey% THEN PROCinitinfo(TRUE)
3760= s_bit% OR i_bit%
3770
3780DEF FNsave
3790IF survey% THEN
3800  filename$ = FNginput(mlx%+96, 1023-16, filename$, 10, 7, 0)
3810  t1% = FNsaveurvey(filepath$, filename$)
3820  = i_bit%
3830ELSE
3840  = 0
3850ENDIF
3860
3870DEF FNmerge
3880IF survey% THEN
3890  filename$ = FNginput(mlx%+96, 1023-16, filename$, 10, 7, 0)
3900  IF FNmergesurvey(filepath$, filename$) THEN PROCinitinfo(FALSE)
3910  = s_bit% OR i_bit% OR p_bit%
3920ELSE
3930  = 0
3940ENDIF
3950
3960DEF FNview
3970REM MOVE 0,0
3971REM MOVE 1279, 1023
3980REM *SGet CaveScreen
3990REM OSCLI("SSave <Survey$Dir>.Screen")
4000view% = Plan
4010= s_bit%
4020
4030DEF FNnorth
4040PROCmouse(2)
4050PROCmouse(1)
4060info%!x_north% = mx%
4070info%!y_north% = my%
4080= s_bit%
4090
4100DEF FNhelp
4110PROChelp("", 12, 13)
4120help_on% = NOT help_on%
4130= i_bit%
4140
4150DEF FNcommand
4160PROCcleardisplay
4170REPEAT
4180  PRINT "*";
4190  $buf% = FNinput(POS, VPOS, "", 255, 12, 7)
4200  SYS "XOS_CLI", buf% TO eptr%,;eflgs%
4210  IF FNTbit(eflgs%, %0001) THEN
4220    t1% = 4
4230    WHILE eptr%?t1% <> 0
4240      VDU eptr%?t1%
4250      t1% += 1

```

```

4260      ENDWHILE
4270      PRINT
4280  ENDIF
4290 UNTIL $buf% = ""
4300 VDU 26
4310 PROCcleardisplay
4320 = s_bit%
4330
4340 DEF FNgrid
4350 grid% = NOT grid%
4360 = s_bit%
4370
4380 DEF FNpaper
4390 IF survey% THEN
4400   papersize% = (papersize%+1) MOD 6
4410   = i_bit%
4420 ELSE
4430   = 0
4440 ENDIF
4450
4460 DEF FNprint
4470 IF survey% THEN
4480   PROCcleardisplay
4490   PRINT TAB(20, 8); "Are you sure (Y/N)?"
4500   a$ = FNinput(40, 8, "Y", 1, 7, 0)
4510   COLOUR 12
4520   COLOUR 128+7
4530   CLS
4540   VDU 26
4550   IF a$ = "Y" OR a$ = "y" THEN
4560     PROChelp("    Printing", 11, 13)
4570     OS_lx% = info%!x_min%
4580     OS_by% = info%!y_min%
4590     wxlim% = OS_Scale%*factor%
4600     wylim% = OS_Scale%*factor%
4610     OS_ScreenX = 7087
4620     OS_ScreenY = 7087
4630     REM Convert survey width to screen and then to inches
4640     t1% = FNwtovx(info%!x_max%)
4650     orgx% = ((p_r%-p_l%)*180/72000-t1%)/2
4660     IF orgx% < 0 THEN orgx% = 0
4670     REM Same for height of survey
4680     t1% = FNwtovy(info%!y_max%)
4690     orgy% = ((p_t%-p_b%-3*72000)*180/72000-t1%)/2
4700     IF orgy% < 0 THEN orgy% = 0
4710     PROCprintpic(printfile$)
4720     PROChelp("", 12, 13)
4730     orgx% = 0
4740     orgy% = 0
4750   ENDIF
4760   = s_bit% OR i_bit%
4770 ELSE
4780   = 0
4790 ENDIF
4800
4810 DEF FNquit
4820 PROCcleardisplay
4830 PRINT TAB(20, 8); "Are you sure (Y/N)?"
4840 a$ = FNinput(40, 8, "Y", 1, 7, 0)
4850 COLOUR 12
4860 COLOUR 128+7
4870 CLS
4880 VDU 26
4890 IF a$ = "Y" OR a$ = "y" THEN
4900   quit% = TRUE
4910   = 0
4920 ELSE
4930   = s_bit% OR i_bit%
4940 ENDIF
4950
4960 DEF FNupleft

```

```

4970= FNup OR FNleft
4980
4990DEF FNupright
5000= FNup OR FNright
5010
5020DEF FNdownleft
5030= FNdown OR FNleft
5040
5050DEF FNdownright
5060= FNdown OR FNright
5070
5080DEF FNleft
5090wlx% -= wxi%
5100wrx% -= wxi%
5110= s_bit%
5120
5130DEF FNright
5140wlx% += wxi%
5150wrx% += wxi%
5160= s_bit%
5170
5180DEF FNup
5190wby% += wyi%
5200wty% += wyi%
5210= s_bit%
5220
5230DEF FNdown
5240wby% -= wyi%
5250wty% -= wyi%
5260= s_bit%
5270
5280DEF FNrotright
5290IF survey% THEN
5300    info%!su_rot% -= 1
5310    IF (info%!su_rot%) < 0 THEN info%!su_rot% = 359
5320    = s_bit% OR i_bit%
5330ELSE
5340    = 0
5350ENDIF
5360
5370DEF FNdegree
5380PROCiconxy("degree", 3, t1%, t2%)
5390t1% = VAL(FNginput(t1%, t2%, STR$(info%!su_rot%), 3, 7, 0))
5400WHILE t1% < 0
5410    t1% += 360
5420ENDWHILE
5430t1% = t1% MOD 360
5440info%!su_rot% = t1%
5450= s_bit% OR i_bit%
5460
5470DEF FNrotleft
5480IF survey% THEN
5490    info%!su_rot% = ((info%!su_rot%)+1) MOD 360
5500    = s_bit% OR i_bit%
5510ELSE
5520    = 0
5530ENDIF
5540
5550DEF FNzoomout
5560IF wlx%+2*wxix% < wrx% AND wby%+2*wyix% < wty% THEN
5570    wlx% += wxix%
5580    wrx% -= wxix%
5590    wby% += wyix%
5600    wty% -= wyix%
5610    wxix% = (wrx%-wllx%)/32
5620    wyix% = (wty%-wby%)/32
5630    info%!su_mag% = 100*(info%!x_max%-info%!x_min%)/(wrx%-wllx%)
5640ENDIF
5650= s_bit% OR i_bit%
5660
5670DEF FNdata1

```

```

5680IF survey% THEN
5690  PROCcleardisplay
5700  PRINT TAB(20, 8);"Are you sure (Y/N)?"
5710  a$ = FNinput(40, 8, "Y", 1, 7, 0)
5720  COLOUR 12
5730  COLOUR 128+7
5740  CLS
5750  VDU 26
5760  IF a$ = "Y" OR a$ = "y" THEN
5770    PROChelp("      Printing", 11, 13)
5780    PROCcleardisplay
5790    VDU 2
5800    PROCbold(TRUE)
5810    PRINT TAB(20);"      Title : ";$(info%+surveysize%+title%)'
5820    PRINT TAB(20);"Surveyed by : ";$(info%+surveysize%+people%)
5830    PRINT TAB(20);"      Date : ";$(info%+surveysize%+date%)
5840    t1% = FNgetsinfo(0, s_x%)
5850    t2% = FNgetsinfo(0, s_y%)
5860    $buf% = FNprintf(FNos(t1%),5,3)+" "+FNprintf(FNos(t2%),5,3)
5870    PRINT TAB(20);"Entrance at : ";$buf%;
5880    PRINT ;" Alt. ";STR$(INT(FNgetsinfo(0, s_z%))); "m"'
5890    PRINT TAB(3);"Station";TAB(15);"N";TAB(25);"E";TAB(35);"Alt"'
5900    FOR s% = 0 TO info%!su_sta%-1
5910      PROCbold(TRUE)
5920      smem% = stations%+s%*stationsize%
5930      PRINT TAB(5);$(smem%+s_name%);
5940      PROCbold(FALSE)
5950      PRINT TAB(15);FNos(FNgetsinfo(s%, s_x%))-t1%;
5960      PRINT TAB(25);FNos(FNgetsinfo(s%, s_y%))-t2%;
5970      PRINT TAB(35);INT(FNgetsinfo(s%, s_z%))
5980    NEXT
5990    PROCeject
6000    VDU 3
6010    VDU 26
6020    PROChelp("", 12, 13)
6030  ENDIF
6040  = s_bit%
6050ELSE
6060  = 0
6070ENDIF
6080
6090DEF FNmagnify
6100IF survey% THEN
6110  PROCiconxy("magnify", 4, t1%, t2%)
6120  t1% = VAL(FNginput(t1%, t2%, STR$(info%!su_mag%), 4, 7, 0))
6130  IF t1% > 0 THEN
6140    PROCinitinfo(FALSE)
6150    info%!su_mag% = t1%
6160    t2% = wrx% - wlx%
6170    adj% = (t2% - 100*t2%/t1%)/2
6180    wlx% += adj%
6190    wrx% -= adj%
6200    t2% = wty% - wby%
6210    adj% = (t2% - 100*t2%/t1%)/2
6220    wby% += adj%
6230    wty% -= adj%
6240  ENDIF
6250  = s_bit% OR i_bit% OR p_bit%
6260ELSE
6270  = 0
6280ENDIF
6290
6300DEF FNzoomin
6310wlx% -= wxi%
6320wrx% += wxi%
6330wby% -= wyi%
6340wty% += wyi%
6350wxi% = (wrx%-wlx%)/32
6360wyi% = (wty%-wby%)/32
6370info%!su_mag% = 100*(info%!x_max%-info%!x_min%)/(wrx%-wlx%)
6380= s_bit% OR i_bit%

```

```

6390
6400DEF FNswap
6410IF survey% THEN
6420    su_no% += 1
6430    IF su_no% > (info%!survey_no%) THEN su_no% = 1
6440    st_no% = 0
6450    st_base% = info!(su_no%*surveysize%+start_sta%)
6460    st_name$ = $(stations%+(st_base%+st_no%)*stationsize%+s_name%)
6470    = s_bit% OR i_bit%
6480ELSE
6490    = 0
6500ENDIF
6510
6520DEF FNnext
6530IF survey% THEN
6540    st_no% += 1
6550    IF st_no% = info!(su_no%*surveysize%+station_no%) THEN st_no% = 0
6560    st_name$ = $(stations%+(st_base%+st_no%)*stationsize%+s_name%)
6570    = s_bit% OR i_bit%
6580ELSE
6590    = 0
6600ENDIF
6610
6620DEF FNleft
6630= FNstation
6640
6650DEF FNright
6660= FNstation
6670
6680DEF FNstation
6690IF survey% THEN
6700    PROCiconxy("sleft", 1, t1%, t2%)
6710    $buf% = FNginput(t1%, t2%, st_name$, 7, 7, 0)
6720    IF FNexists($buf%, t1%) THEN
6730        st_no% = t1%
6740        st_name$ = $buf%
6750        t2% = 0
6760        su_no% = 1
6770        WHILE t2% < t1%
6780            t2% += info!(su_no%*surveysize%+station_no%)
6790            su_no% += 1
6800        ENDWHILE
6810        su_no% -= 1
6820        st_base% = info!(su_no%*surveysize%+start_sta%)
6830        st_no% -= st_base%
6840    ENDIF
6850    = s_bit% OR i_bit%
6860ELSE
6870    = 0
6880ENDIF
6890
6900DEF FNprev
6910IF survey% THEN
6920    st_no% -= 1
6930    IF st_no% < 0 THEN st_no% = (info!(su_no%*surveysize%+station_no%))-1
6940    st_name$ = $(stations%+(st_base%+st_no%)*stationsize%+s_name%)
6950    = s_bit% OR i_bit%
6960ELSE
6970    = 0
6980ENDIF
6990
7000DEF FNblank
7010= 0
7020
7030DEF PROCshow(st_no%, wlx%, wby%, wrx%, wty%, flags%)
7040REM f% are the redraw flags
7050REM b0 - everything
7060REM b1 - redraw survey
7070REM b2 - rewrite station info
7080REM b3 - rewrite position info
7090wxlim% = wrx%-wlx%

```

```

7100wylim% = wty%-wby%
7110OS_ScreenX = vrx%-vlx%
7120OS_ScreenY = vty%-vby%
7130OS_lx% = wlx%
7140OS_by% = wby%
7150IF Fntbit(flags%, e_bit% OR s_bit%) THEN
7160   bank% = bank% EOR 3
7170   SYS "OS_Byte", 112, bank%
7180ENDIF
7190IF Fntbit(flags%, e_bit%) THEN
7200   PROCarea(vlx%, vby%, vrx%, vty%, 7, 15)
7210   PROCarea(mlx%, mby%, mrx%, mty%, 13, 15)
7220   PROCarea(vlx%, vty%, mrx%, 1023, 8, 15)
7230   PROCicons
7240   GCOL 15
7250   RECTANGLE 0, 0, 1279, 1023
7260   RECTANGLE vlx%, vty%, 34*16, 1023-vty%
7270   RECTANGLE vlx%+34*16, vty%, 9*16, 1023-vty%
7280   RECTANGLE vlx%+43*16, vty%, 9*16, 1023-vty%
7290   RECTANGLE mlx%+iw%, mty%, 3*iw%, 1023-mty%
7300ENDIF
7310IF Fntbit(flags%, s_bit%) AND survey% THEN
7320   PROCsetblock(blk%, vlx%+2, vby%+4, vrx%-2, vty%-4)
7330   PROCshowsurvey(blk%, FALSE)
7340   t1% = st_base%+st_no%
7350   LINE pt%(t1%,0)-32, pt%(t1%,1), pt%(t1%,0)+32, pt%(t1%,1)
7360   LINE pt%(t1%,0), pt%(t1%,1)-32, pt%(t1%,0), pt%(t1%,1)+32
7370   CIRCLE pt%(t1%,0), pt%(t1%,1), 24
7380   VDU 26
7390ENDIF
7400IF Fntbit(flags%, i_bit%) AND survey% THEN
7410   smem% = info%+su_no%*surveysize%
7420   PROCupdate(vlx%+16, vty%+48, $(smem%+title%), 33, 12, 8)
7430   $buf% = STR$(INT((smem%!survey_len%)/factor%)) + "m"
7440   PROCupdate(vlx%+16+34*16, vty%+48, $buf%, 7, 12, 8)
7450   $buf% = STR$(INT((info%!total_len%)/factor%)) + "m"
7460   PROCupdate(vlx%+16+43*16, vty%+48, $buf%, 7, 12, 8)
7470   PROCiconxy("paper", 2, t1%, t2%)
7480   PROCupdate(t1%, t2%, "A"+STR$papersize%, 2, 7, 0)
7490   PROCiconxy("degree", 3, t1%, t2%)
7500   PROCupdate(t1%, t2%, FNprintf(info%!su_rot%,3,0), 3, 12, 13)
7510   PROCiconxy("magnify", 4, t1%, t2%)
7520   PROCupdate(t1%, t2%, FNprintf(info%!su_mag%,4,0), 4, 12, 13)
7530   smem% = stations%+(st_base%+st_no%)*stationsize%
7540   t1% = FNgetsinfo(st_base%, s_x%)-(smem%!s_x%)/factor%
7550   t2% = FNgetsinfo(st_base%, s_y%)-(smem%!s_y%)/factor%
7560   $buf% = "N "+STR$(t1%)+ " E "+STR$(t2%)
7570   PROCupdate(vlx%+16+52*16, vty%+48, $buf%, 10, 12, 8)
7580   PROCiconxy("sleft", 1, t1%, t2%)
7590   PROCupdate(t1%, t2%, st_name$, 7, 12, 13)
7600   PROCupdate(mlx%+96, 1023-16, filename$, 10, 12, 8)
7610   t1% = (smem%!s_x%)/factor%
7620   t2% = (smem%!s_y%)/factor%
7630   PROCosref(33, t1%, t2%, "Station ")
7640   PROCiconpos(41, 13, t1%, t2%)
7650   PROCupdate(t1%, t2%, "OS Scale 1:"+STR$(OS_Scale%), 18, 12, 13)
7660ENDIF
7670IF Fntbit(flags%, p_bit%) AND survey% THEN PROCmousepos(mx%, my%)
7680IF Fntbit(flags%, e_bit% OR s_bit%) THEN
7690   SYS "OS_Byte", 113, bank%
7700   WAIT
7710ENDIF
7720ENDPROC
7730
7740DEF PROCcleardisplay
7750GCOL 12
7760GCOL 128+7
7770VDU 24, vlx%+2; vby%+4; vrx%-2; vty%-4;
7780CLG
7790COLOUR 12
7800COLOUR 128+7

```

```

7810VDU 28, vlx%DIV16+1, (1023-vby%DIV32-1, vrx%DIV16-1, (1023-vty%DIV32+1
7820CLS
7830ENDPROC
7840
7850DEF PROCosref(i%, x%, y%, i$)
7860PROCiconpos(i%, 13, t1%, t2%)
7870$buf% = FNprintf(FNos(x%),5,3)+" "+FNprintf(FNos(y%),5,3)
7880PROCupdate(t1%-8, t2%, i$+$buf%, 19, 12, 13)
7890ENDPROC
7900
7910DEF FNos(o%)
7920REM Make o% into a 100km map reference
7930IF o% >= 100000 THEN o% = o% MOD 100000
7940WHILE o% < 0
7950    o% += 100000
7960ENDWHILE
7970= o%
7980
7990DEF PROCnorth
8000LOCAL x%, y%, s, c
8010s = sin(info%!su_rot%)
8020c = cos(info%!su_rot%)
8030x% = info%!x_north%+orgx%
8040y% = info%!y_north%+orgy%
8050FOR i% = 1 TO Npts%
8060    t1% = north%(i%,0)
8070    t2% = north%(i%,1)
8080    MOVE x%+t1%*c-t2%*s, y%+t1%*s+t2%*c
8090    t1% = north%(i%,2)
8100    t2% = north%(i%,3)
8110    DRAW x%+t1%*c-t2%*s, y%+t1%*s+t2%*c
8120NEXT
8130ENDPROC
8140
8150DEF PROCicons
8160LOCAL ipx%, ipy%
8170ipx% = ix%
8180ipy% = iy%-ih%
8190FOR i% = 0 TO icons%
8200    PROCicon(icon$(i%, 0), ipx%, ipy%)
8210    ipx% += iw%
8220    IF ipx% > mrx%-iw% THEN
8230        ipx% = ix%
8240        ipy% -= ih%
8250    ENDIF
8260NEXT
8270ipx% = ix%
8280PROCicon("infobox", ipx%, ipy%)
8290ipy% -= ih%
8300PROCicon("infobox", ipx%, ipy%)
8310ipy% -= ih%
8320PROCicon("infobox", ipx%, ipy%)
8330ipy% -= ih%
8340PROCicon("infobox", ipx%, ipy%)
8350ENDPROC
8360
8370DEF PROCicon(name$, x%, y%)
8380MOVE x%, y%
8390$buf% = name$
8400SYS "Wimp_SpriteOp", 512+24,, buf% TO ,, t1%
8410SYS "Wimp_SpriteOp", 512+28,,t1%
8420ENDPROC
8430
8440DEF PROCiconxy(icon$, fw%, RETURN x%, RETURN y%)
8450LOCAL i%
8460i% = 0
8470WHILE icon$ <> icon$(i%, 0)
8480    i% += 1
8490ENDWHILE
8500PROCiconpos(i%, fw%, x%, y%)
8510ENDPROC

```

```

8520
8530DEF PROCiconpos(i%, fw%, RETURN x%, RETURN y%)
8540x% = ix% + (i% MOD 4)*iw% + (iw% DIV 2) - (fw%*16 DIV 2)
8550y% = iy% - (i% DIV 4)*ih% - (ih% DIV 2) + 12
8560ENDPROC
8570
8580DEF PROCshowsurvey(blk%, print%)
8590VDU 24, blk%!0; blk%!4; blk%!8; blk%!12;
8600IF print% THEN
8610    GCOL 7
8620    GCOL 128+0
8630ELSE
8640    GCOL 13
8650    GCOL 128+7
8660    CLG
8670ENDIF
8680old_to = -1
8690old_b = 0
8700old_w = 0
8710tx% = FNvtowx((vrx%+vlx%) DIV 2)
8720ty% = FNvtowy((vty%+vby%) DIV 2)
8730FOR s% = 1 TO (info%!survey_no%)
8740    nv% = info%!(s%*surveysize%+vector_no%)
8750    bv% = info%!(s%*surveysize%+start_vec%)
8760    bs% = info%!(s%*surveysize%+start_sta%)
8770    FOR v% = 0 TO nv%-1
8780        vmem% = vectors%+(bv%+v%)*vectorsize%
8790        from% = bs%+(vmem%!v_from%)/factor%
8800        to% = bs%+(vmem%!v_to%)/factor%
8810        smem% = stations%+from%*stationsize%
8820        b = (vmem%!v_bearing%)/factor%
8830        IF v% = 0 THEN
8840            REM calc with leg bearing and width of from station only
8850            PROCcalc(from%, smem%, b)
8860        ELSE
8870            IF from% = old_to THEN
8880                REM continuing passage
8890                REM calc with average of previous and this leg bearing
8900                REM & width of from station only
8910                PROCcalc(from%, smem%, FNav(b, old_b))
8920            ELSE
8930                REM terminate old bit of survey
8940                REM calc with bearing and to station width
8950                PROCcalc(old_to, stations%+old_to*stationsize%, old_b)
8960            ENDIF
8970        ENDIF
8980        old_to = to%
8990        old_b = b
9000        old_w = w
9010    NEXT
9020    REM finish last bit
9030    PROCcalc(old_to, stations%+old_to*stationsize%, old_b)
9040NEXT
9050FOR s% = 1 TO (info%!survey_no%)
9060    nv% = info%!(s%*surveysize%+vector_no%)
9070    bv% = info%!(s%*surveysize%+start_vec%)
9080    bs% = info%!(s%*surveysize%+start_sta%)
9090    IF NOT print% THEN
9100        IF s% = su_no% THEN
9110            GCOL 10
9120        ELSE
9130            GCOL 12
9140        ENDIF
9150    ENDIF
9160    FOR v% = 0 TO nv%-1
9170        vmem% = vectors%+(bv%+v%)*vectorsize%
9180        from% = bs%+(vmem%!v_from%)/factor%
9190        to% = bs%+(vmem%!v_to%)/factor%
9200        MOVE pt%(from%,0)-pt%(from%,2)+orgx%,pt%(from%,1)-pt%(from%,3)+orgy%
9210        DRAW pt%(to%, 0)-pt%(to%,2)+orgx%, pt%(to%,1)-pt%(to%,3)+orgy%
9220        PLOT85,pt%(from%,0)+pt%(from%,2)+orgx%,pt%(from%,1)+pt%(from%,3)+orgy%

```

```

9230    PLOT 85, pt%(to%,0)+pt%(to%,2)+orgx%, pt%(to%, 1)+pt%(to%, 3)+orgy%
9240    NEXT
9250NEXT
9260IF print% THEN
9270    t3% = 0
9280    PROCouts(300, 1620, "          Title : "+$(info%+surveysize%+title%))
9290    PROCouts(300, 1580, "Surveyed by : "+$(info%+surveysize%+people%))
9300    PROCouts(300, 1540, "          Date : "+$(info%+surveysize%+date%))
9310    t1% = FNgetsinfo(0, s_x%)
9320    t2% = FNgetsinfo(0, s_y%)
9330    $buf% = FNprintf(FNos(t1%),5,3)+" "+FNprintf(FNos(t2%),5,3)
9340    $buf% += "    Alt. "+STR$(INT(FNgetsinfo(0, s_z%)))+ "m"
9350    PROCouts(300, 1500, "Entrance at : "+$buf%)
9360    t1% = (info%!total_len%)/factor%
9370    PROCouts(300, 1460, "          Length : "+STR$(t1%)+ "m")
9380    PROCouts(300, 1420, "          Scale : 1:" +STR$(OS_Scale%))
9390    RECTANGLE 260, 1348, t3%*16+5*16, 320
9400ENDIF
9410IF NOT print% THEN GCOL 9
9420PROCnorth
9430ENDPROC
9440
9450DEF PROCouts(x%, y%, s$)
9460IF LEN(s$) > t3% THEN t3% = LEN(s$)
9470MOVE x%, y%
9480PRINT s$
9490ENDPROC
9500
9510DEF PROCcalc(s%, smem%, b)
9520t1% = (smem%!s_x%)-tx%
9530t2% = (smem%!s_y%)-ty%
9540pt%(s%, 0) = FNwtovx(t1%*cos(info%!su_rot%)-t2%*sin(info%!su_rot%)+tx%)
9550pt%(s%, 1) = FNwtovy(t1%*sin(info%!su_rot%)+t2%*cos(info%!su_rot%)+ty%)
9560w = (smem%!s_w%)*959/wxlim%
9570pt%(s%, 2) = w*SIN(RAD(b-90-info%!su_rot%))/2
9580pt%(s%, 3) = w*COS(RAD(b-90-info%!su_rot%))/2
9590ENDPROC
9600
9610DEF FNav(b1, b2)
9620LOCAL av
9630IF b1 > 180 THEN b1 -= 360
9640IF b2 > 180 THEN b2 -= 360
9650av = (b1+b2)/2
9660IF av >= 360 THEN av -= 360
9670= av
9680
9690DEF FNloadsurvey(path$, name$)
9700IF FNfile(&ff, path$+name$+".Coords", stations%, 0,0,0) THEN
9710    IF FNfile(&ff, path$+name$+".Vectors", vectors%, 0,0,0) THEN
9720        = FNfile(&ff, path$+name$+".Info", info%, 0,0,0)
9730    ENDIF
9740ENDIF
9750PROCerror(eptr%)
9760= FALSE
9770
9780DEF FNsavesurvey(path$, name$)
9790IF FNfile(&00, path$+name$+".Coords", 0,0, stations%, stations%+stationsize%*(info%!su
|_sta%)) THEN
9800    IF FNfile(&00, path$+name$+".Vectors", 0,0, vectors%, vectors%+vectorsize%*(info%!s
|u_vec%)) THEN
9810        = FNfile(&00, path$+name$+".Info", 0,0, info%, info%+surveysize%*(info%!survey_n
|o%+1))
9820    ENDIF
9830ENDIF
9840PROCerror(eptr%)
9850= FALSE
9860
9870DEF FNmergesurvey(path$, name$)
9880LOCAL pos%, lim%, ns%, ts%, tv%
9890pos% = (info%!su_sta%)*stationsize%
9900lim% = maxstations%*stationsize%

```

```

9910IF FNfload(path$+name$+".Coords", stations%, pos%, lim%) THEN
9920  pos% = (info%!su_vec%)*vectorsize%
9930  lim% = maxvectors%*vectorsize%
9940  IF FNfload(path$+name$+".Vectors", vectors%, pos%, lim%) THEN
9950    t1% = OPENIN(path$+name$+".Info")
9960    IF t1% THEN
9970      ns% = FNgetw(t1%)
9980      IF ns%+(info%!survey_no%) < maxsurveys% THEN
9990        t2% = FNgetw(t1%) : t2% = FNgetw(t1%)
10000        t2% = FNgetw(t1%) : t2% = FNgetw(t1%)
10010        ts% = FNgetw(t1%)
10020        tv% = FNgetw(t1%)
10030        t2% = FNgetw(t1%): IF t2%<info%!x_min% THEN info%!x_min% = t2%
10040        t2% = FNgetw(t1%): IF t2%<info%!y_min% THEN info%!y_min% = t2%
10050        t2% = FNgetw(t1%): IF t2%<info%!z_min% THEN info%!z_min% = t2%
10060        t2% = FNgetw(t1%): IF t2%>info%!x_max% THEN info%!x_max% = t2%
10070        t2% = FNgetw(t1%): IF t2%>info%!y_max% THEN info%!y_max% = t2%
10080        t2% = FNgetw(t1%): IF t2%>info%!z_max% THEN info%!z_max% = t2%
10090        info%!total_len% += FNgetw(t1%)
10100        PROCgets(t1%, info%+total_len%+4, surveysize%-(total_len%+4))
10110        FOR s% = 1 TO ns%
10120          smem% = info%+(info%!survey_no%+s%)*surveysize%
10130          smem%!station_no% = FNgetw(t1%)
10140          smem%!vector_no% = FNgetw(t1%)
10150          smem%!start_sta% = FNgetw(t1%) + info%!su_sta%
10160          smem%!start_vec% = FNgetw(t1%) + info%!su_vec%
10170          smem%!survey_len% = FNgetw(t1%)
10180          PROCgets(t1%, smem%+title%, 50)
10190          PROCgets(t1%, smem%+people%, 50)
10200          PROCgets(t1%, smem%+date%, 50)
10210        NEXT
10220        info%!su_sta% += ts%
10230        info%!su_vec% += tv%
10240        info%!survey_no% += ns%
10250        CLOSE#t1%
10260        = TRUE
10270      ELSE
10280        CLOSE#t1%
10290      ENDIF
10300    ENDIF
10310  ENDIF
10320ENDIF
10330= FALSE
10340
10350DEF FNfile(fop%, $buf%, f2%, f3%, f4%, f5%)
10360SYS "XOS_File", fop%, buf%, f2%, f3%, f4%, f5% TO eptr%;eflgs%
10370= NOT FNTbit(eflgs%, %0001)
10380
10390DEF FNfsize(fname$)
10400t1% = OPENIN(fname$)
10410IF t1% <> 0 THEN
10420  t2% = EXT#t1%
10430  CLOSE#t1%
10440  = t2%
10450ELSE
10460  = 1E6 :REM Large value prevents loading
10470ENDIF
10480
10490DEF FNfload(fname$, mem%, pos%, lim%)
10500IF FNfsize(fname$)+pos% < lim% THEN
10510  = FNfile(&ff, fname$, mem%+pos%, 0,0,0)
10520ELSE
10530  = FALSE
10540ENDIF
10550
10560DEF FNgetw(f%)
10570= (BGET#f%) + (BGET#f% << 8) + (BGET#f% << 16) + (BGET#f% << 24)
10580
10590DEF PROCgets(f%, mem%, l%)
10600LOCAL i%
10610FOR i% = 0 TO l%-1

```

```

10620 mem%i% = BGET#f%
10630NEXT
10640ENDPROC
10650
10660DEF PROCreadsurvey(path$, name$)
10670REM Reading a survey craps out any currently loaded surveys, so beware
10680file$ = path$ + name$ + "." + name$
10690PROCinitsurvey
10700COLOUR 12
10710COLOUR 135
10720end% = FALSE
10730error% = FALSE
10740PROCnewfile(file$)
10750WHILE NOT (error% OR end%)
10760 CASE FNreaditem_s OF
10770     WHEN "NOTES" : PRINT FNnewline
10780     WHEN "TITLE" : $(info%+survey%+title%) = LEFT$(FNnewline, 49)
10790     WHEN "PEOPLE" : $(info%+survey%+people%) = LEFT$(FNnewline, 49)
10800     WHEN "DATE" : $(info%+survey%+date%) = LEFT$(FNnewline, 49)
10810     WHEN "ORIGIN" : PROCorigin
10820     WHEN "MOLE" : PROCmolephone
10830     WHEN "UNITS" : PROCunits
10840     WHEN "STYLE" : PROCstyle
10850     WHEN "GAUGE" : PROCgauge
10860     WHEN "CHANGE" : PROCchange
10870     WHEN "FORMAT" : PROCformat
10880     WHEN "FILE" : PROCnewfile(FNreaditem_s)
10890     WHEN "END" : PROCendfile
10900     WHEN "" : PRINT FNnewline
10910     OTHERWISE : PROCsurveyleg
10920 ENDCASE
10930ENDWHILE
10940PRINT 'info%!su_sta%;" survey stations read"
10950IF SUM(handle%()) <> 0 THEN
10960 PROCreport(104)
10970 CLOSE#0
10980ENDIF
10990survey% = NOT error%
11000IF survey% THEN PROCinitinfo(TRUE)
11010PROCcontinue
11020ENDPROC
11030
11040DEF PROCinitinfo(new%)
11050wlx% = info%!x_min%
11060wby% = info%!y_min%
11070wrx% = info%!x_max%
11080wty% = info%!y_max%
11090adj% = ((wrx%-wlx%)-(wty%-wby%)) DIV 2
11100IF adj% > 0 THEN
11110 wby% -= adj%
11120 wty% += adj%
11130ELSE
11140 wlx% += adj%
11150 wrx% -= adj%
11160ENDIF
11170wxi% = (wrx%-wlx%)/32
11180wyi% = (wty%-wby%)/32
11190IF new% THEN
11200 info%!survey_no% = 1
11210 PROCtotal
11220 su_no% = 1
11230 st_name$ = $(stations%+s_name%)
11240ENDIF
11250ENDPROC
11260
11270DEF PROCorigin
11280REM Define new origin
11290PRINT FNnewline :REM print any comment
11300name$ = FNreaditem_s :REM origin name
11310IF FNexists(name$, t1%) THEN
11320 PROCreport(4)

```

```

11330ELSE
11340    REM Read info about origin
11350    x = FNreaditem_n(Units)      :REM read 10 figure map reference
11360    y = FNreaditem_n(Units)
11370    z = FNreaditem_n(Units)      :REM height above sea level
11380    w = FNreaditem_n(Units)
11390    h = FNreaditem_n(Units)
11400    PROCaddstation(name$, x, y, z, w, h, or_bit%)
11410    PRINT FNnewline
11420ENDIF
11430ENDPROC
11440
11450DEF PROCmolephone
11460PRINT FNnewline
11470name$ = FNreaditem_s :REM Mole Phone location
11480from$ = FNreaditem_s :REM Station where correction is to start
11490IF FNexists(name$, t1%) AND FNexists(from$, t2%) THEN
11500    IF old_to$ = name$ THEN
11510        x = FNreaditem_n(Units)      :REM read 10 figure map reference
11520        y = FNreaditem_n(Units)
11530        z = FNreaditem_n(Units)      :REM height above sea level!, not depth
11540        tx = FNgetsinfo(t1%, s_x%)
11550        ty = FNgetsinfo(t1%, s_y%)
11560        tz = FNgetsinfo(t1%, s_z%)
11570        COLOUR 9
11580        PRINT"MOLE PHONE location for ";name$;" correction starts at ";from$
11590        PRINT "Correction is:"
11600        PRINT " x : "; x-tx
11610        PRINT " y : "; y-ty
11620        PRINT " z : "; z-tz
11630
11640        lptr% = 0
11650        cur_sta% = t1%
11660        v% = 0
11670        WHILE cur_sta% <> t2% AND (FNgetsinfo(cur_sta%, s_flags%) < 16) AND v% <> -1
11680            slist%(lptr%, 0) = cur_sta%
11690            lptr% += 1
11700            cur_sta% = FNfrom(cur_sta%, v%)
11710        ENDWHILE
11720        slist%(lptr%, 0) = cur_sta%
11730
11740        ex = (x-tx)/lptr%
11750        ey = (y-ty)/lptr%
11760        ez = (z-tz)/lptr%
11770
11780        IF slist%(lptr%, 0) <> t2% THEN PROCreport(105)
11790
11800        COLOUR 9
11810        PRINT "Adjusting : ";
11820        FOR i% = lptr% TO 1 STEP -1
11830            t1% = FNfindvector(slist%(i%,0), slist%(i%-1,0))
11840            PROCgetvector(t1%, from%, to%, l, b, i, dx, dy, dz)
11850            dx += ex
11860            dy += ey
11870            dz += ez
11880            l = SQR(dx*dx + dy*dy + dz*dz)
11890            b = FNbearing(l, dx, dy)
11900            i = FNbearing(l, dy, dz)
11910            PROCputvector(t1%, from%, to%, l, b, i, dx, dy, dz)
11920            PROCgetstation(from%, x, y, z)
11930            PROCputstation(to%, x+dx, y+dy, z+dz)
11940            PROCputsinfo(to%, s_flags%, FNgetsinfo(to%, s_flags%) OR vz_bit%)
11950            PRINT $(stations%+to%*stationsize%+s_name%);" ";
11960        NEXT
11970
11980        lptr% = 0
11990        FOR i% = 0 TO info%!su_sta%-1
12000            j% = FNgetsinfo(i%, s_flags%)
12010            IF FNTbit(j%, vz_bit%) AND FNgetsinfo(i%, s_fromcount%) > 1 THEN
12020                REM Add junction to list with successors
12030                v% = 0

```

```

12040         REPEAT
12050             t1% = FNfindto(i%, v%)
12060             IF t1% <> -1 AND (FNgetsinfo(t1%, s_flags%) < 64) THEN
12070                 slist%(lptr%, 0) = i%
12080                 slist%(lptr%, 1) = t1%
12090                 lptr% += 1
12100             ENDIF
12110         UNTIL t1% = -1
12120     ENDIF
12130 NEXT
12140
12150     bptr% = 0
12160     WHILE lptr% > bptr%
12170         t1% = FNfindvector(slist%(bptr%,0), slist%(bptr%,1))
12180         v% = 0
12190         REPEAT
12200             PROCgetvector(t1%, from%, to%, l, b, i, dx, dy, dz)
12210             t2% = FNTbit(FNgetsinfo(to%, s_flags%), vz_bit%)
12220             IF NOT t2% THEN
12230                 PROCgetstation(from%, x, y, z)
12240                 PROCputstation(to%, x+dx, y+dy, z+dz)
12250                 PROCputsinfo(to%,s_flags%,FNgetsinfo(to%,s_flags%) OR vz_bit%)
12260                 PRINT $(stations%+to%*stationsize%+s_name%);" ";
12270             ENDIF
12280             t1% = FNfindto(to%, v%)
12290             IF FNgetsinfo(to%, s_tocount%) > 1 AND NOT t2% THEN
12300                 j% = v%
12310                 i% = FNfindto(to%, j%)
12320                 WHILE i% <> -1
12330                     IF FNgetsinfo(i%, s_flags%) < 64 THEN
12340                         slist%(lptr%, 0) = to%
12350                         slist%(lptr%, 1) = i%
12360                         lptr% += 1
12370                     ENDIF
12380                     i% = FNfindto(to%, j%)
12390                 ENDWHILE
12400             ENDIF
12410             UNTIL t1% <> -1 OR t2%
12420             bptr% += 1
12430         ENDWHILE
12440
12450         COLOUR 12
12460         PROCputsinfo(slist%(0, 0), s_flags%, mo_bit%)
12470         PROCtotal
12480         PRINT FNnewline
12490     ELSE
12500         PROCreport(7)
12510     ENDIF
12520 ELSE
12530     PROCreport(2)
12540 ENDIF
12550 ENDPROC
12560
12570 DEF FNfindvector(from%, to%)
12580 v% = 0
12590 WHILE (FNgetvinfo(v%,v_from%)<>from% OR FNgetvinfo(v%,v_to%)<>to%) AND (v%<info%!su_ve
|c%)
12600     v% += 1
12610 ENDWHILE
12620 = v%
12630
12640 DEF FNfindto(from%, RETURN v%)
12650 LOCAL tmp%
12660 WHILE (FNgetvinfo(v%, v_from%) <> from%) AND (v% < info%!su_vec%)
12670     v% += 1
12680 ENDWHILE
12690 IF v% < (info%!su_vec%) THEN
12700     tmp% = FNgetvinfo(v%, v_to%)
12710     v% += 1
12720     = tmp%
12730 ELSE

```

```

12740     = -1
12750ENDIF
12760
12770DEF FNfrom(to%, RETURN v%)
12780LOCAL v%, i%, j%, k%
12790REM Given a 'to' station, find vector and return the 'from' station
12800REM If there is more than one, then find the non-closure route
12810i% = FNgetsinfo(to%, s_tocount%)
12820IF i% > 1 THEN
12830    j% = 0
12840    REPEAT
12850        k% = FNfindfrom(to%, v%)
12860        j% += 1
12870    UNTIL (NOT FNTbit(FNgetsinfo(k%, s_flags%), cl_bit%)) OR i%=j%
12880    IF FNTbit(FNgetsinfo(k%, s_flags%), cl_bit%) THEN PROCreport(9)
12890    = k%
12900ELSE
12910    = FNfindfrom(to%, v%)
12920ENDIF
12930
12940DEF FNfindfrom(to%, RETURN v%)
12950LOCAL tmp%
12960WHILE (FNgetvinfo(v%, v_to%) <> to%) AND (v% < info%!su_vec%)
12970    v% += 1
12980ENDWHILE
12990IF v% < (info%!su_vec%) THEN
13000    tmp% = FNgetvinfo(v%, v_from%)
13010    v% += 1
13020    = tmp%
13030ELSE
13040    = -1
13050ENDIF
13060
13070DEF PROCunits
13080CASE FNreaditem_s OF
13090    WHEN "METRIC"
13100        unit_factor = 1.0
13110    WHEN "FEET"
13120        unit_factor = 0.3048
13130    WHEN "CONVERT"
13140        unit_factor = FNreaditem_n(Raw)
13150    OTHERWISE
13160        PROCreport(102)
13170        unit_factor = 1.0
13180ENDCASE
13190PRINT FNnewline
13200ENDPROC
13210
13220DEF PROCstyle
13230CASE FNreaditem_s OF
13240    WHEN "NORMAL"
13250        style% = Normal
13260    WHEN "DIVING"
13270        REM Diving must be accompanied by a survey station and initial depth
13280        PRINT FNnewline
13290        IF NOT FNexists(FNreaditem_s, t1%) THEN PROCreport(0)
13300        depth = FNreaditem_n(Units)
13310        style% = Diving
13320    WHEN "UPDOWN"
13330        style% = UpDown
13340    OTHERWISE
13350        PROCreport(100)
13360        style% = Normal
13370ENDCASE
13380PRINT FNnewline
13390ENDPROC
13400
13410DEF PROCgauge
13420REM Read adjustments for depth gauge readings
13430surface_depth = FNreaditem_n(Units)
13440gauge_factor = FNreaditem_n(Units)

```

```

13450PRINT FNnewline
13460ENDPROC
13470
13480DEF PROCchange
13490REM Read adjustments for compass and clinometer readings
13500compass_adj = FNreaditem_n(Units)
13510clino_adj   = FNreaditem_n(Units)
13520PRINT FNnewline
13530ENDPROC
13540
13550DEF PROCformat
13560CASE FNreaditem_s OF
13570     WHEN "FREE"
13580         format% = Free
13590     WHEN "FIXED"
13600         format% = Fixed
13610     OTHERWISE
13620         PROCreport(103)
13630         format% = Free
13640ENDCASE
13650PRINT FNnewline
13660ENDPROC
13670
13680DEF PROCnewfile(file$)
13690files% += 1
13700IF files% > maxfiles% THEN
13710     PROCreport(101)
13720ELSE
13730     handle%(files%) = OPENIN(file$)
13740     file$(files%) = file$
13750     IF handle%(files%) = 0 THEN
13760         PROCreport(3)
13770         files% -= 1
13780     ELSE
13790         PRINT FNnewline
13800     ENDIF
13810ENDIF
13820ENDPROC
13830
13840DEF PROCendfile
13850PRINT "Closing file : ";file$(files%)
13860CLOSE#handle%(files%)
13870handle%(files%) = 0
13880files% -= 1
13890IF files% > 0 THEN
13900     PRINT "Continue input from "; file$(files%); " (Y/N) ";
13910     REPEAT a$ = GET$ : UNTIL INSTR("YyNn", a$)
13920     IF a$ = "N" OR a$ = "n" THEN end% = TRUE
13930ELSE
13940     end% = TRUE
13950ENDIF
13960ENDPROC
13970
13980DEF PROCsurveyleg
13990LOCAL to$, from$, pl, l, b, i, dz, closure%
14000from$ = $buf%
14010to$ = FNreaditem_s
14020l = FNreaditem_n(Units)           :REM length
14030b = FNreaditem_n(Raw)-compass_adj :REM bearing
14040CASE style% OF
14050     WHEN Normal
14060         i = FNreaditem_n(Raw)-clino_adj           :REM incline
14070     WHEN Diving
14080         i = (FNreaditem_n(Units)-surface_depth)*gauge_factor :REM abs. depth
14090     WHEN UpDown
14100         i = FNreaditem_n(Units)                   :REM rel. depth
14110ENDCASE
14120closure% = FALSE
14130CASE FNexists(from$, t1%)+FNexists(to$, t2%) OF
14140     WHEN 0 :REM both stations unknown
14150         PROCreport(1)

```

```

14160 WHEN -2 :REM both stations are known
14170     closure% = TRUE
14180     IF to$ = old_to$ THEN
14190         REM Swap stations if out of order
14200         SWAP from$, to$
14210         SWAP t1%, t2%
14220         b = (b+180) MOD 360 :REM Reverse bearing
14230         IF style% <> Diving THEN i = -i :REM and negate inclination
14240     ENDIF
14250     PROCputsinfo(t1%, s_fromcount%, FNgetsinfo(t1%, s_fromcount%)+1)
14260     PROCputsinfo(t2%, s_tocount%, FNgetsinfo(t1%, s_tocount%)+1)
14270 WHEN -1
14280     IF t1% = (info%!su_sta%) THEN
14290         REM Swap stations if out of order
14300         SWAP from$, to$
14310         SWAP t1%, t2%
14320         b = (b+180) MOD 360 :REM Reverse bearing
14330         IF style% <> Diving THEN i = -i :REM and negate inclination
14340     ENDIF
14350     PROCputsinfo(t1%, s_fromcount%, FNgetsinfo(t1%, s_fromcount%)+1)
14360 ENDCASE
14370 x = FNgetsinfo(t1%, s_x%)
14380 y = FNgetsinfo(t1%, s_y%)
14390 z = FNgetsinfo(t1%, s_z%)
14400 CASE style% OF
14410     WHEN Normal
14420         REM Convert inclination to get vertical displacement
14430         dz = l*SIN(RAD(i))
14440     WHEN Diving
14450         REM Calculate relative depth change and inclination
14460         dz = depth-i
14470         depth = i
14480         i = DEG(ASN(dz/l))
14490     WHEN UpDown
14500         REM Calculate inclination from given vertical displacement
14510         dz = i
14520         i = DEG(ASN(dz/l))
14530 ENDCASE
14540 p1 = SQR(l*l - dz*dz)
14550 dx = p1 * SIN(RAD(b))
14560 dy = p1 * COS(RAD(b))
14570 w = FNreaditem_n(Units)
14580 h = FNreaditem_n(Units)
14590 IF closure% THEN
14600     PROCaddvector(t1%, t2%, l, b, i, dx, dy, dz)
14610     VDU 7
14620     COLOUR 9
14630     PRINT '"Loop Closure !!"'
14640     tx = FNgetsinfo(t2%, s_x%)
14650     ty = FNgetsinfo(t2%, s_y%)
14660     tz = FNgetsinfo(t2%, s_z%)
14670     ex = tx-(x+dx)
14680     ey = ty-(y+dy)
14690     ez = tz-(z+dz)
14700     PRINT "This passage survey is out by:"
14710     PRINT " x : ";ex;"m"
14720     PRINT " y : ";ey;"m"
14730     PRINT " z : ";ez;"m"
14740     REM leg$ holds the start station
14750     REM vec% holds the start vector number
14760     REM leg% is the number of stations in this run minus 1
14770     ex = ex/(leg%+1)
14780     ey = ey/(leg%+1)
14790     ez = ez/(leg%+1)
14800     PRINT "Adjusting : ";
14810     PROCgetvector(vec%, from%, to%, l, b, i, dx, dy, dz)
14820     REPEAT
14830         dx += ex
14840         dy += ey
14850         dz += ez
14860         l = SQR(dx*dx + dy*dy + dz*dz)

```

```

14870      b = FNbearing(1, dx, dy)
14880      i = FNbearing(1, dy, dz)
14890      PROCputvector(vec%, from%, to%, 1, b, i, dx, dy, dz)
14900      PROCgetstation(from%, x, y, z)
14910      PROCputstation(to%, x+dx, y+dy, z+dz)
14920      PROCputsinfo(to%, s_flags%, FNgetsinfo(to%, s_flags%) OR cl_bit%)
14930      PRINT $(stations%+to%*stationsize%+s_name%);" ";
14940      vec% += 1
14950      PROCgetvector(vec%, from%, to%, 1, b, i, dx, dy, dz)
14960      UNTIL to% = t2%
14970      COLOUR 12
14980ELSE
14990      PROCaddstation(to$, x+dx, y+dy, z+dz, w, h, 0)
15000      PROCaddvector(t1%, t2%, 1, b, i, dx, dy, dz)
15010      PROCputsinfo(t2%, s_tocount%, FNgetsinfo(t1%, s_tocount%)+1)
15020ENDIF
15030IF from$ <> old_to$ THEN
15040      leg$ = from$
15050      leg% = 1
15060      vec% = (info%!su_vec%)-1
15070ELSE
15080      leg% += 1
15090ENDIF
15100old_to$ = to$
15110PRINT FNnewline
15120ENDPROC
15130
15140DEF FNbearing(1, dx, dy)
15150LOCAL b
15160b = DEG(ACS(dy/1))
15170IF dx < 0 THEN b = 360-b
15180= b
15190
15200DEF FNreaditem_n(cflags%)
15210LOCAL tmp
15220tmp = VAL(FNreaditem_s)
15230CASE cflags% OF
15240      WHEN Raw
15250          = tmp
15260      WHEN Units
15270          = tmp*unit_factor
15280      WHEN Internal
15290          = tmp*unit_factor*factor%
15300ENDCASE
15310
15320DEF FNreaditem_s
15330LOCAL char%, c%
15340$buf% = ""
15350c% = 0
15360PROCskip(32)
15370REPEAT
15380      char% = FNchar
15390      buf%?c% = char%
15400      c% += 1
15410UNTIL char% <= 32
15420REM Allow empty fields to be processed
15430IF char% <> 32 THEN PTR#handle%(files%) = PTR#handle%(files%) - 1
15440buf%?(c%-1) = CR
15450= $buf%
15460
15470DEF FNconvert_n(number)
15480REM Convert from current units only
15490REM unit_factor converts the units to metric
15500= number*unit_factor
15510
15520DEF FNnewline
15530LOCAL char%, s$
15540REPEAT
15550      char% = FNchar
15560      s$ += CHR$(char%)
15570UNTIL char% < 32

```

```

15580= s$
15590
15600DEF PROCskip(char%)
15610REPEAT UNTIL FNchar <> char%
15620IF PTR#handle%(files%) <> 0 THEN PTR#handle%(files%)=PTR#handle%(files%)-1
15630ENDPROC
15640
15650DEF FNchar
15660LOCAL char%
15670IF EOF#handle%(files%) THEN
15680    end% = TRUE
15690    char% = CR
15700ELSE
15710    char% = BGET#handle%(files%)
15720ENDIF
15730IF char% < 32 THEN char% = CR
15740= char%
15750
15760DEF FNgetsinfo(s%, s_off%)
15770= (stations%!(s%*stationsize% + s_off%))/factor%
15780
15790DEF PROCputsinfo(s%, s_off%, d%)
15800stations%!(s%*stationsize% + s_off%) = d%*factor%
15810ENDPROC
15820
15830DEF FNgetvinfo(v%, v_off%)
15840= (vectors%!(v%*vectorsize% + v_off%))/factor%
15850
15860DEF PROCgetstation(s%, RETURN x, RETURN y, RETURN z)
15870mem% = stations%+s%*stationsize%
15880x = (mem%!s_x%)/factor%
15890y = (mem%!s_y%)/factor%
15900z = (mem%!s_z%)/factor%
15910ENDPROC
15920
15930DEF PROCputstation(s%, x, y, z)
15940mem% = stations%+s%*stationsize%
15950mem%!s_x% = x*factor%
15960mem%!s_y% = y*factor%
15970mem%!s_z% = z*factor%
15980ENDPROC
15990
16000DEF PROCaddstation(name$, x, y, z, w, h, flags%)
16010LOCAL x%, y%, z%
16020IF (info%!su_sta%) = maxstations% THEN PROCreport(5)
16030x% = x*factor%
16040y% = y*factor%
16050z% = z*factor%
16060mem% = stations%+(info%!su_sta%)*stationsize%
16070PRINT name$; TAB(10);x;TAB(25);y;TAB(40);z;
16080$(mem%+s_name%) = name$
16090mem%!s_tocount% = 0
16100mem%!s_fromcount% = 0
16110mem%!s_flags% = flags%
16120mem%!s_x% = x%
16130mem%!s_y% = y%
16140mem%!s_z% = z%
16150mem%!s_w% = w*factor%
16160mem%!s_h% = h*factor%
16170info%!su_sta% += 1
16180info%!(surveysize%+station_no%) += 1
16190IF x% > info%!x_max% THEN info%!x_max% = x%
16200IF x% < info%!x_min% THEN info%!x_min% = x%
16210IF y% > info%!y_max% THEN info%!y_max% = y%
16220IF y% < info%!y_min% THEN info%!y_min% = y%
16230IF z% > info%!z_max% THEN info%!z_max% = z%
16240IF z% < info%!z_min% THEN info%!z_min% = z%
16250ENDPROC
16260
16270DEF PROCgetvector(v%, RETURN f%, RETURN t%, RETURN l, RETURN b, RETURN i,
|         RETURN dx, RETURN dy, RETURN dz)

```

```

16280mem% = vectors%+v%*vectorsize%
16290f% = (mem%!v_from%)/factor%
16300t% = (mem%!v_to%)/factor%
16310l = (mem%!v_length%)/factor%
16320b = (mem%!v_bearing%)/factor%
16330i = (mem%!v_incline%)/factor%
16340dx = (mem%!v_dx%)/factor%
16350dy = (mem%!v_dy%)/factor%
16360dz = (mem%!v_dz%)/factor%
16370ENDPROC
16380
16390DEF PROCputvector(v%, from%, to%, l, b, i, dx, dy, dz)
16400mem% = vectors%+v%*vectorsize%
16410mem%!v_from% = from%*factor%
16420mem%!v_to% = to%*factor%
16430mem%!v_length% = l*factor%
16440mem%!v_bearing% = b*factor%
16450mem%!v_incline% = i*factor%
16460mem%!v_dx% = dx*factor%
16470mem%!v_dy% = dy*factor%
16480mem%!v_dz% = dz*factor%
16490ENDPROC
16500
16510DEF PROCaddvector(from%, to%, l, b, i, dx, dy, dz)
16520IF (info!su_vec%) = maxvectors% THEN PROCreport(6)
16530PROCputvector(info!su_vec%, from%, to%, l, b, i, dx, dy, dz)
16540info!su_vec% += 1
16550info!(surveysize%+vector_no%) += 1
16560ENDPROC
16570
16580DEF PROCtotal
16590LOCAL t
16600FOR t1% = 0 TO (info!su_vec%)-1
16610 t += FNgetvinfo(t1%, v_length%)
16620NEXT
16630info!total_len% = t*factor%
16640info!(surveysize%+survey_len%) = t*factor%
16650ENDPROC
16660
16670DEF FNexists(station$, RETURN s%)
16680LOCAL found%
16690REM Search for named station, returns TRUE/FALSE and index s%
16700REM If s% > info!su_sta% THEN the station was not found
16710s% = 0
16720found% = FALSE
16730WHILE (s% < (info!su_sta%)) AND (NOT found%)
16740 IF $(stations%+s%*stationsize%+s_name%) = station$ THEN
16750 found% = TRUE
16760 ELSE
16770 s% += 1
16780 ENDIF
16790ENDWHILE
16800= found%
16810
16820DEF PROCreport(ern%)
16830PRINT '
16840VDU 7
16850CASE ern% OF
16860 WHEN 0 : PRINT "Missing or undefined survey station for DIVING"
16870 WHEN 1 : PRINT "Both stations undefined, have you specified an ORIGIN"
16880 WHEN 2 : PRINT "Survey station unknown for MOLE PHONE location"
16890 WHEN 3 : PRINT "Can't find input file"
16900 WHEN 4 : PRINT "Origin is re-defining existing station!"
16910 WHEN 5 : PRINT "Too many survey stations"
16920 WHEN 6 : PRINT "Too many survey legs"
16930 WHEN 7 : PRINT "MOLE station must follow leg data"
16940 WHEN 8 : PRINT "Can't find station for MOLE correction"
16950 WHEN 9 : PRINT "Cannot find un-closed leg for MOLE correction"
16960 WHEN 100 : PRINT "Missing or unknown survey STYLE"
16970 WHEN 101 : PRINT "File nesting too deep, ignoring data"
16980 WHEN 102 : PRINT "Unknown UNITS, metric assumed"

```

```

16990 WHEN 103 : PRINT "Unknown FORMAT, free format assumed"
17000 WHEN 104 : PRINT "Missing END keyword"
17010 WHEN 105 : PRINT "Can't reach destination station for MOLE correction"
17020ENDCASE
17030IF ern% < 100 THEN
17040 COLOUR 11
17050 PRINT '"Fatal error occurred on input"'
17060 COLOUR 12
17070 @% = &90A
17080 CLOSE#0
17090 error% = TRUE
17100ENDIF
17110PRINT '
17120ENDPROC
17130
17140DEF PROCcontinue
17150PRINT "Press MOUSE to continue"
17160PROCmouse(3)
17170ENDPROC
17180
17190DEF PROCmouse(flq%)
17200*Fx 21 9
17210omx% = mx%
17220omy% = my%
17230omb% = mb%
17240IF Fntbit(flq%, 1) THEN REPEAT MOUSE mx%,my%,mb%:UNTIL mb% > 0
17250IF Fntbit(flq%, 2) THEN REPEAT MOUSE mx%,my%,mb%:UNTIL mb% = 0
17260IF Fntbit(flq%, 4) THEN REPEAT MOUSE mx%,my%,mb%:UNTILmx%<>omx%ORmy%<>omy%
17270MOUSE mx%, my%, mb%
17280ENDPROC
17290
17300DEF PROCmousepos(x%, y%)
17310PROCosref(37, FNvtowx(mx%)/factor%, FNvtowy(my%)/factor%, "OS Ref: ")
17320ENDPROC
17330
17340DEF PROCarea(lx%, by%, rx%, ty%, c1%, c2%)
17350GCOL c1%
17360RECTANGLE FILL lx%, by%, rx%-lx%, ty%-by%
17370GCOL c2%
17380RECTANGLE lx%, by%, rx%-lx%, ty%-by%
17390ENDPROC
17400
17410DEF PROCupdate(x%, y%, s$, fw%, fc%, bc%)
17420FOR b% = 1 TO 2
17430 SYS "OS_Byte", 112, b%
17440 PROCgprint(x%, y%, STRING$(fw%, CHR$(255)), bc%)
17450 PROCgprint(x%, y%, s$, fc%)
17460NEXT
17470SYS "OS_Byte", 112, bank%
17480ENDPROC
17490
17500DEF PROCgprint(x%, y%, s$, c%)
17510VDU 5
17520GCOL c%
17530MOVE x%, y%
17540PRINT s$;
17550VDU 4
17560OFF
17570ENDPROC
17580
17590DEF FNprintf(v, fw%, fl%)
17600REM Print value v in a field width of fw%
17610REM fl% bit 0 = 0 left justify = 1 right justify
17620REM bit 1 = 0 pad with spaces = 1 pad with zeros
17630IF Fntbit(fl%, 2) THEN
17640 p$ = STRING$(fw%, "0")
17650ELSE
17660 p$ = STRING$(fw%, " ")
17670ENDIF
17680IF Fntbit(fl%, 1) THEN
17690 = RIGHT$(p$+STR$v, fw%)

```

```

17700ELSE
17710  = LEFT$(STR$V+p$, fw%)
17720ENDIF
17730
17740DEF FNinput(x%, y%, d$, l%, fc%, bc%)
17750ON
17760*Fx 21 0
17770COLOUR fc%
17780COLOUR 128+bc%
17790PRINT TAB(x%, y%);d$;
17800REPEAT
17810  k% = INKEY(0)
17820  CASE k% OF
17830    WHEN 127
17840      IF d$ <> "" THEN d$ = LEFT$(d$, LEN(d$)-1)
17850      PRINT TAB(x%+LEN(d$), y%);" ";
17860    WHEN 21
17870      PRINT TAB(x%, y%);STRING$(LEN(d$), " ");
17880      d$ = ""
17890    OTHERWISE
17900      IF k% > 31 AND k% < 127 AND LEN(d$) < l% THEN d$ += CHR$(k%)
17910  ENDCASE
17920  IF k% > 0 THEN PRINT TAB(x%, y%);d$;
17930UNTIL k% = 13
17940PRINT
17950OFF
17960= d$
17970
17980DEF FNginput(x%, y%, d$, l%, fc%, bc%)
17990ON
18000*Fx 21 0
18010PROCupdate(x%, y%, d$, l%, fc%, bc%)
18020REPEAT
18030  k% = INKEY(0)
18040  CASE k% OF
18050    WHEN 127
18060      IF d$ <> "" THEN d$ = LEFT$(d$, LEN(d$)-1)
18070    WHEN 21
18080      d$ = ""
18090    OTHERWISE
18100      IF k% > 31 AND k% < 127 AND LEN(d$) < l% THEN d$ += CHR$(k%)
18110  ENDCASE
18120  IF k% > 0 THEN
18130    PROCupdate(x%, y%, d$, l%, fc%, bc%)
18140  ENDIF
18150UNTIL k% = 13
18160OFF
18170= d$
18180
18190DEF FNTbit(d%, mask%)
18200= (d% AND mask%) > 0
18210
18220DEF FNwtovx(x%)
18230= (x%-OS_lx%)*OS_ScreenX/wxlim%
18240
18250DEF FNwtovy(y%)
18260= (y%-OS_by%)*OS_ScreenY/wylim%
18270
18280DEF FNvtowx(x%)
18290= INT(x%*wxlim%/OS_ScreenX+wlx%)
18300
18310DEF FNvtowy(y%)
18320= INT(y%*wylim%/OS_ScreenY+wby%)
18330
18340DEF PROCtrig
18350DIM sin(359), cos(359)
18360FOR a = 0 TO 359
18370  sin(a) = SIN(RAD(a))
18380  cos(a) = COS(RAD(a))
18390NEXT
18400ENDPROC

```

```

18410
18420DEF PROCprintpic(file$)
18430file% = OPENOUT(file$)
18440LOCAL e%, e$
18450LOCAL ERROR
18460ON ERROR LOCAL : RESTORE ERROR : PROCrderr : CLOSE#file% : PROCerr
18470SYS selectjob%, file% TO old%
18480LOCAL ERROR
18490ON ERROR LOCAL : RESTORE ERROR : PROCrderr : SYS abortjob%, file% :      SYS selectjo
|b%, old% : PROCerr
18500SYS giverect%, 1, wnd%, mat%, org%, &ffffff00
18510SYS drawpage%, 1, blk%, 0, 0 TO copies%, id%
18520WHILE copies% <> 0
18530  PROCshowsurvey(blk%, TRUE)
18540  SYS getrect%, blk% TO copies%, id%
18550ENDWHILE
18560SYS endjob%, file%
18570RESTORE ERROR
18580SYS selectjob%, old%
18590RESTORE ERROR
18600CLOSE #file%
18610ENDPROC
18620
18630DEF PROCsetblock(blk%, p1%, p2%, p3%, p4%)
18640REM Set up parameter block
18650blk%!0 = p1%
18660blk%!4 = p2%
18670blk%!8 = p3%
18680blk%!12 = p4%
18690ENDPROC
18700
18710DEF PROCrderr
18720e% = ERR
18730e$ = REPORT$
18740ENDPROC
18750
18760DEF PROCerr
18770CLOSE #0
18780MODE 12
18790ERROR e%, e$ + "(from line " + STR$(ERL) + ")"
18800ENDPROC
18810
18820DEF PROCerror(eptr%)
18830t1% = 4
18840WHILE eptr%?t1% <> 0 AND t1% < 33
18850  buf%?(t1%-4) = eptr%?t1%
18860  t1% += 1
18870ENDWHILE
18880buf%?(t1%-4) = 13
18890PROCupdate(vlx%+16, vty%+48, $buf%, 33, 0, 8)
18900VDU 7
18910TIME = 0
18920REPEAT UNTIL TIME > 200
18930ENDPROC
18940
18950DEF PROCshowinfo
18960PRINT "Surveys : ";info%!survey_no%
18970PRINT "Stations: ";info%!su_sta%
18980PRINT "Vectors : ";info%!su_vec%
18990FOR s% = 1 TO info%!survey_no%
19000  mem% = info%+s%*surveysize%
19010  PRINT "Stations : ";mem%!station_no%
19020  PRINT "Vectors : ";mem%!vector_no%
19030  PRINT "Start Sta: ";mem%!start_sta%
19040  PRINT "Start Vec: ";mem%!start_vec%
19050  PRINT "Length : ";mem%!survey_len%
19060  PRINT "Title : ";$(mem%+title%)
19070  PRINT "People : ";$(mem%+people%)
19080  PRINT "Date : ";$(mem%+date%)
19090  PRINT "Scale : ";$(mem%+OS_Scale%)
19100NEXT

```

```

19110ENDPROC
19120
19130DEF PROCshowstations
19140FOR s% = 0 TO info%!su_sta%-1
19150     mem% = stations%+s%*stationsize%
19160     PRINT s%
19170     PRINT "x      : ";mem%!s_x%
19180     PRINT "y      : ";mem%!s_y%
19190     PRINT "z      : ";mem%!s_z%
19200     PRINT "w      : ";mem%!s_w%
19210     PRINT "h      : ";mem%!s_h%
19220     PRINT "name : ";$(mem%+s_name%)
19230NEXT
19240ENDPROC
19250
19260DEF PROCbold(flag%)
19270IF flag% THEN
19280     VDU 1,27,1,ASC"G"
19290ELSE
19300     VDU 1,27,1,ASC"H"
19310ENDIF
19320ENDPROC
19330
19340DEF PROCcentre(flag%)
19350IF flag% THEN
19360     VDU 1,27,1,ASC"a",1,ASC"1"
19370ELSE
19380     VDU 1,27,1,ASC"a",1,ASC"0"
19390ENDIF
19400ENDPROC
19410
19420DEF PROCdw(flag%)
19430IF flag% THEN
19440     VDU 1,27,1,ASC"w",1,ASC"1",1,27,1,ASC"W",1,ASC"1"
19450ELSE
19460     VDU 1,27,1,ASC"w",1,ASC"0",1,27,1,ASC"W",1,ASC"0"
19470ENDIF
19480ENDPROC
19490
19500DEF PROCeject
19510VDU 1, 12
19520ENDPROC

```